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D76 - ITERATIVE DESIGN AND USABILITY TESTING AND DEVELOPMENT OF CASEYCHAT: A LLM-POWERED CONVERSATIONAL AGENT FOR INDIVIDUALS EXPERIENCING HOMELESSNESS

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Background: Unstable housing is linked to adverse mental health outcomes. Barriers such as stigma and fragmented services hinder care. The Kearney Center (KC), a shelter in Tallahassee, faces high demand and case management burdens. CaseyChat, a large language model (LLM)-powered chatbot, aimed to help clients navigate resources and services. Our prior needs assessments (Phase 1) highlighted misalignments between client needs, service access, and staff capacity at KC. Building on the findings, Phase 2 focused on iterative design, usability testing, and refinement of CaseyChat from prototypes to a minimum viable product (MVP).

Methods: Phase 2 involved five steps: 1) Participants (n = 7) engaged in a co-design session to provide feedback on app flow and features using a high-fidelity prototype; 2) Usability testing (n = 4) informed interface improvements and backend development; 3) Participants (n = 6) in co-design sessions ranked CaseyChat features and matched them with needs; 4) Usability testing (n = 5) evaluated app navigation and feature clarity, leading to refinements of frontend functionality; 5) A beta MVP was tested (n = 7) through one-on-one interviews and a two-week field trial. Staff (n = 6) provided operational perspectives and validated design decisions through focus groups.

Results: Nineteen priority features were identified, with strongest demand for information services, tracking, and app onboarding. Usability testing revealed issues such as non-clickable elements, inaccurate service information, and unclear framing of the “daily check-in.” Iterative design added anonymous sign-in, service cards, app customization (e.g., font), and navigation cues. MVP testing showed user acceptance: participants described CaseyChat as intuitive, supportive, and empathetic, referring to the chatbot as human-like. Challenges persisted, including incomplete service information and lack of conversation memory. Staff emphasized CaseyChat’s potential to ease case management burdens, enhance mental health support, and streamline workflow, while noting the importance of multilingual access and real-time accuracy. Discussion &

Conclusion: Phase 2 demonstrated that iterative steps testing enhanced CaseyChat’s design and usability as a supportive tool for individuals experiencing homelessness. Next step is a feasibility study (Phase 3) to evaluate CaseyChat’s feasibility and acceptability and potential for integration into existing service workflows.

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D65 - KEEP CALM AND DROP OUT? PREDICTING ATTRITION IN AN MHEALTH MEDITATION APP USING BAYESIAN MODEL AVERAGED LOGISTIC REGRESSION

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Background: Despite the proven growing evidence of the benefits of mindfulness meditation delivered via mobile health (mHealth) apps, high attrition remains a critical barrier. Meditation apps account for a majority of active users of mental health apps, but less than five percent of initial users continue practice after 30 days. Some evidence has uncovered demographics and personality factors that predict attrition for traditional in-person formats, yet the reasons why participants drop out of app-based meditation remain unclear.

Objective: This project applies Bayesian model-averaged logistic regression to capture dropout patterns and identify profiles of users most at risk for attrition. Method: The sample of 1,390 participants (82% non-male; age M = 35.6 years, SD = 10.4) completed self-report measures of stress, burnout, sleep quality, negative affect, food insecurity, and adverse childhood experiences. A Bayesian model-averaged logistic regression was conducted to evaluate all possible combinations of ten theoretically relevant predictors (1,024 candidate models) and to average the predictive effects across them.

Results: The full model including ten predictors had the highest posterior model probability (0.285), although no single model dominated the model space. The Bayes factor (BF) on the model odds indicated that the odds of favoring this model increased by a factor of 2.2. In terms of parameter inclusion, the strongest predictor of attrition at the follow-up was condition (inclusion BF = 20.911), with participants in the meditation group being about 38% more likely to drop out. Income also emerged as a meaningful predictor (inclusion BF = 3.258): middle-income, high-income, and “prefer-not-to-say” participants were approximately 36–39% less likely to drop out than low-income participants. Additional predictors with moderate evidence included baseline stress and adverse childhood experiences, with inclusion BFs ranging from 2.7 to 3.2. Each was associated with small differences in dropout likelihood (approximately –2% and +3%, respectively).

Discussion: Findings indicate specific predictors of attrition in an app-based mindfulness meditation intervention with the factors of condition and income being most meaningful. These results have utility for informing targeted strategies to improve engagement and laying the foundation for larger-scale interventions that enhance the effectiveness and reach of mHealth mindfulness programs.

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